43

| Version No. | | | | |
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| 3 | 0 | 0 | 8 | 1 |
| ⓪ | ● | ● | ⓪ | ⓪ |
| ① | ① | ① | ① | ● |
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| ● | ③ | ③ | ③ | ③ |
| ④ | ④ | ④ | ④ | ④ |
| ⑤ | ⑤ | ⑤ | ⑤ | ⑤ |
| ⑥ | ⑥ | ⑥ | ⑥ | ⑥ |
| ⑦ | ⑦ | ⑦ | ⑦ | ⑦ |
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| ③ | ③ | ③ | ③ | ③ | ③ | ③ |
| ④ | ④ | ④ | ④ | ④ | ④ | ④ |
| ⑤ | ⑤ | ⑤ | ⑤ | ⑤ | ⑤ | ⑤ |
| ⑥ | ⑥ | ⑥ | ⑥ | ⑥ | ⑥ | ⑥ |
| ⑦ | ⑦ | ⑦ | ⑦ | ⑦ | ⑦ | ⑦ |
| ⑧ | ⑧ | ⑧ | ⑧ | ⑧ | ⑧ | ⑧ |
| ⑨ | ⑨ | ⑨ | ⑨ | ⑨ | ⑨ | ⑨ |

# MATHEMATICS HSSC-I

## SECTION – A (Marks 20)

Time allowed: 25 Minutes

**Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed.**

*Do not use lead pencil.*

حصہ اوّل لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

**Answer Sheet No.** _____

ہر سوال کے سامنے دیے گئے، کریکولم کے مطابق درست دائرہ کو پُر کریں۔

**Invigilator Sign.** _____

**Fill the relevant bubble against each question according to curriculum:**

**Candidate Sign.** _____

| | Question | A | B | C | D | A | B | C | D |
|---|---|---|---|---|---|---|---|---|---|
| 1. | What are the multiplicative factors of $z^2+4z+5$? | $(z+2-i)$ $(z+2+i)$ | $(z-2-i)$ $(z-2+i)$ | $(z+2-2i)$ $(z+2+2i)$ | $(z-2-2i)$ $(z-2+2i)$ | ○ | ○ | ○ | ○ |
| 2. | The principal argument $(\theta)$ of the complex number $z=-1+i\sqrt{3}$ is: | $-\frac{\pi}{3}$ | $\frac{\pi}{3}$ | $\frac{2\pi}{3}$ | $\frac{5\pi}{3}$ | ○ | ○ | ○ | ○ |
| 3. | A system of linear equations is said to be inconsistent if it has: | No solution | At least one solution | Infinitely many solutions | Exactly two solutions | ○ | ○ | ○ | ○ |
| 4. | If A is a 3 x 3 matrix, then $\lvert 3A\rvert$ results | $3\lvert A\rvert$ | $6\lvert A\rvert$ | $9\lvert A\rvert$ | $27\lvert A\rvert$ | ○ | ○ | ○ | ○ |
| 5. | In an arithmetic sequence, $a_1=5$ and $a_{15}=75$. What is the common difference? | 3 | 4 | 5 | 6 | ○ | ○ | ○ | ○ |
| 6. | If $\frac{1}{2}, \frac{1}{x}, \frac{1}{5}$ are in harmonic sequence then what is the value of $x$? | $\frac{2}{7}$ | $\frac{7}{2}$ | $\frac{7}{10}$ | $\frac{10}{7}$ | ○ | ○ | ○ | ○ |
| 7. | What is the first term of an infinite geometric series whose sum is 10, and common ratio 0.6? | 4 | 5 | 6 | 7 | ○ | ○ | ○ | ○ |
| 8. | What is the sum of the series up to 4 terms? $1+3+5+\cdots+(2n-1)=n^2 \quad \forall n\in N$ | 7 | 9 | 15 | 16 | ○ | ○ | ○ | ○ |
| 9. | What is the middle term in the expansion of $(x+y)^6$? | $\binom{6}{2}x^3y^3$ | $\binom{6}{3}x^3y^3$ | $\binom{6}{3}x^4y^2$ | $\binom{6}{2}x^4y^2$ | ○ | ○ | ○ | ○ |
| 10. | In signal processing, $P(x)$ represents a signal. If $P(-1)=0$, then the signal has: | A zero at $x=-1$ | Unstablity | Periodicity | Error | ○ | ○ | ○ | ○ |
| 11. | If dot product of two vectors is zero then angle between them is: | 180° | 90° | 45° | 0° | ○ | ○ | ○ | ○ |

| | Question | A | B | C | D | A | B | C | D |
|---|---|---|---|---|---|---|---|---|---|
| 12. | If $\vec{a}.\vec{b}\times\vec{c}=12$ then volume of parallelepiped with $\vec{a}$ , $\vec{b}$ and $\vec{c}$ as co–terminal edges is: | 6 | 12 | 24 | 36 | ○ | ○ | ○ | ○ |
| 13. | A force $\vec{F}=2\hat{i}+\hat{j}+3\hat{k}$ moves an object along a displacement vector $\vec{d}=3\hat{i}+4\hat{j}-2\hat{k}$. What is the work done? | –4 | 10 | 4 | 16 | ○ | ○ | ○ | ○ |
| 14. | If $\tan\theta=2$, what is the value of $\mathrm{Cos}\,\theta$ when θ is in the first quadrant? | $\frac{1}{\sqrt{3}}$ | $\frac{2}{\sqrt{5}}$ | $\sqrt{5}$ | $\frac{1}{\sqrt{5}}$ | ○ | ○ | ○ | ○ |
| 15. | If $sinx=\frac{3}{5}$, what is the value of $sin2x$ ? $(0°<x<90°)$ | $\frac{12}{25}$ | $\frac{24}{25}$ | $\frac{7}{25}$ | $\frac{25}{25}$ | ○ | ○ | ○ | ○ |
| 16. | If graph of a function is symmetric about origin (e.g. $\mathrm{Sin}\,x$ ), what type of function is it? | Even | Odd | Neither Even nor Odd | Periodic but not Symmetric | ○ | ○ | ○ | ○ |
| 17. | The range of a trigonometric function $4sin4x$ is: | $[-1, 1]$ | $[-4, 4]$ | $[-16, 16]$ | $(-\infty,+\infty)$ | ○ | ○ | ○ | ○ |
| 18. | The expression $sin75^{\circ}cos15^{\circ}$ is equivalent to: | $\frac{1}{2}(\sin 90^{\circ}+\sin 60^{\circ})$ | $\frac{1}{2}(\cos 90^{\circ}+\cos 60^{\circ})$ | $\frac{1}{2}(\sin 60^{\circ}-\sin 90^{\circ})$ | $\frac{1}{2}(\cos 60^{\circ}-\cos 90^{\circ})$ | ○ | ○ | ○ | ○ |
| 19. | How many 5-letter words can be formed from the word 'MATHS' without repeating any letter? | 20 | 60 | 120 | 240 | ○ | ○ | ○ | ○ |
| 20. | In how many different ways a player can select 6 numbers from a set of 49 numbers, in a lottery? | ${}^{49}C_6$ | $\frac{49!}{6!}$ | ${}^{49}P_6$ | $6!$ | ○ | ○ | ○ | ○ |

——1HA-I 25008- 30081(B)——

$$S_{\infty}=\frac{a_1}{1-r}$$

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